

Bio-Plex Pro™ Precision Human Cytokine Assays 11-Plex

Instruction Manual

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Section 1

Introduction

Bio-Plex Pro™ Precision Human Cytokine Assays are highly sensitive magnetic bead-based multiplex assays that allow the accurate measurement of low levels of cytokines in diverse matrices, including serum, plasma, and culture supernatant. The multiplexing feature makes it possible to quantitate the level of multiple cytokines in a single well of a 96-well microplate in just 3 hours, using as little as 12.5 µl of serum or plasma, or 50 µl of culture supernatant.

These assays are offered in a convenient kit format that includes assay, reagent, and diluent components in a single box. Standard diluent for serum and plasma are included, as are additional lyophilized standards and quality control vials.

Visit us on the Web at [bio-rad.com/web/bio-plex](https://www.bio-rad.com/web/bio-plex) for a current listing of Bio-Plex Pro Precision Human Cytokine Assays.

Section 2

Principle

Technology

The Bio-Plex® Multiplex Immunoassay System is built around three core technologies. The first is a novel technology that uses up to 100 unique fluorescently dyed beads (xMAP technology) that permit the simultaneous detection of up to 100 different types of molecules in a single well of a 96-well microplate. The second is a flow cytometer with two lasers and associated optics to measure the different molecules bound to the surface of the beads. The third is a high-speed digital signal processor that efficiently manages the fluorescent output.

Assay Format

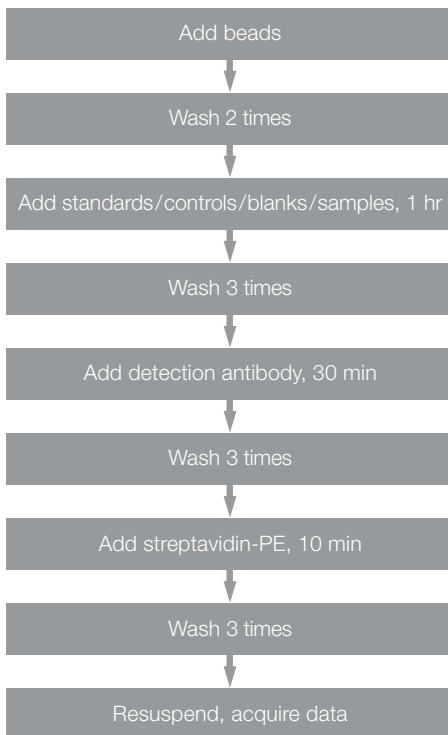
The principle of these 96-well-plate-formatted, bead-based assays is similar to that of a capture sandwich immunoassay. An antibody directed against the desired target cytokine is covalently coupled to internally dyed beads. The coupled beads are allowed to react with a sample containing the target cytokine. After a series of washes to remove unbound protein, a biotinylated detection antibody specific for a different epitope is added to the reaction. The result is the formation of a sandwich of antibodies around the target cytokine. Streptavidin-phycoerythrin (streptavidin-PE) is then added to bind to the biotinylated detection antibodies on the bead surface.

Data Acquisition and Analysis

Data from the reaction are then acquired using the Bio-Plex Suspension Array System (or Luminex System), a dual-laser, flow-based microplate reader system. The contents of the well are drawn up into the reader. The lasers and associated optics detect the internal fluorescence of the individual dyed beads as well as the fluorescent signal on the bead surface. This identifies each assay and reports the level of target protein in the well. Intensity of fluorescence detected on the beads indicates the relative

quantity of targeted molecules. A high-speed digital processor efficiently manages the data output, which is further analyzed and presented as fluorescence intensity on Bio-Plex Manager™ Software, the accompanying software package.

Assay Workflow



Section 3

Required Materials

Bio-Plex Pro™ Precision Human Cytokine Assays are offered in a convenient kit format that includes assay, reagent, and diluent components all in a single box. Analyzing these assays requires the use of Bio-Plex Manager™ Software version 4.1 or higher.

Kit contents.

Component	Units	Volume	Description
Coupled magnetic beads (20x)	1	240 µl	0.5 ml, amber vial
Detection antibodies (1x)	1	Lyophilized	5 ml glass, red crimp
Standard	2	Lyophilized	2 ml glass, silver crimp
Control	1	Lyophilized	2 ml glass, red crimp
Standard diluent	2	Lyophilized	5 ml glass, green crimp
Sample diluent	1	15 ml	60 ml bottle
Assay buffer (5x)		50 ml	125 ml bottle
Streptavidin-PE (100x)	1	75 µl	0.5 ml, amber vial
Flat bottom plate (96-well)	1		
Plate seals	1		1 pack of 4

Storage and Stability

Kit components should be stored at 4°C and should never be frozen. Coupled magnetic beads and streptavidin-PE should be stored in the dark. All components are guaranteed for up to 6 months from the date of purchase when stored as specified in this manual.

Section 4

Recommended Materials

For optimal results, the use of the items below is recommended.

Recommended materials.

Item	Ordering Information
Bio-Plex® Multiplex Immunoassay System (or Luminex System)	Bio-Rad catalog #171000205
Bio-Plex Validation Kit	Bio-Rad catalog #171203001
Bio-Plex Calibration Kit	Bio-Rad catalog #171203060
Microtiter Plate Shaker	
IKA-Schuttler MTS-4 Shaker for 4 microplates or Lab-Line Model 4625 Plate Shaker (or equivalent, capable of 300–1,100 rpm)	IKA catalog #3208000 VWR catalog #57019-600
Vortexer	
VWR brand mini-vortexer	VWR catalog #58816-121
Scientific Instruments Vortex-Genie 2 Mixer	VWR catalog #58815-234
Reagent Reservoirs	Bio-Rad catalog #2244872
Box of 40 packages, 5 reservoirs per pkg (200 reservoirs), 50 ml capacity, graduated, polystyrene	
Tubes and Plates	
Titertube® Micro Test Tubes, pkg of 10 racks, 96 micro test tubes per rack (960 tubes), 1 ml, 8.8 x 45 mm, presterilized, not autoclavable	Bio-Rad catalog #2239395
Micro Test Tubes, pkg of 500, standard micro test tube, capped, 1.5 ml, autoclavable, natural	Bio-Rad catalog #2239501
Bio-Plex Pro™ Flat Bottom Plates, pkg of 40, 96-well plates, for use with Bio-Plex Pro wash stations when using magnetic bead-based assays	Bio-Rad catalog #171025001
Other Materials	
Pipets and pipet tips, 1.5 ml microcentrifuge tubes, 15 ml culture tubes, sterile distilled water, aluminum foil, absorbent paper towels	

Section 5

Sample Preparation

This section provides instructions for preparing samples derived from serum, plasma, and culture supernatant. For sample preparations not mentioned here, consult the publications listed in [Bio-Rad bulletin 5297](#), available for download at bio-rad.com.

Serum and Plasma Samples

Note that for plasma samples, EDTA tubes are recommended; however, sodium citrate tubes are acceptable. Extremely lipemic samples may be filtered with a 0.22 µm filter to prevent clogging. Hemolyzed samples are not suitable for Bio-Plex Pro™ Precision Human Cytokine Assays.

1. Collect and process the serum or plasma samples and assay immediately or freeze at –20°C. Avoid repeat freezing and thawing.
2. Centrifuge the samples at 1,000 rpm for 15 min at 4°C to clear the samples of precipitate. Alternatively, carefully filter the samples with a 0.22 µm filter to prevent instrument clogging.
3. Immediately dilute 1 volume of sample with 3 volumes of sample diluent. Keep the samples on ice until ready for use.

Culture Supernatant Samples

1. Collect and process the culture supernatant samples and assay immediately or freeze at –20°C. Avoid repeat freezing and thawing.
2. If required, dilute the culture supernatant with culture medium; recommended range is Neat to 1:10. Serum-free culture medium should contain carrier protein (such as BSA) at a concentration of at least 0.5%. Keep the samples on ice until ready for use.

Section 6

Standard Preparation

Two tubes of lyophilized cytokine standard are provided in each Bio-Plex Pro™ Precision Human Cytokine Assay. However, only one of the tubes is required per 96-well plate. The product data sheet provided with the assay lists the concentration of the reconstituted standard. This procedure will prepare enough standard to run each dilution in duplicate.

Reconstitute Standards

1. Gently tap the glass vial containing the lyophilized cytokine standard on a solid surface to ensure the pellet is at the bottom.
2. Reconstitute 1 vial of lyophilized standard with 200 μ l of dH₂O.
3. Allow vial to sit at room temperature for 5 min.
4. Gently vortex for 1–3 sec and incubate on ice for 30 min. Be consistent with the incubation time for optimal assay performance.

Prepare Standard Dilution Series

The cytokine concentrations specified for the 8-point standard dilution set have been selected for optimized curve fitting using the 5-parameter logistic (5PL) or 4-parameter logistic (4PL) regression in Bio-Plex Manager™ Software. Results generated using dilution points other than those listed in this manual have not been optimized.

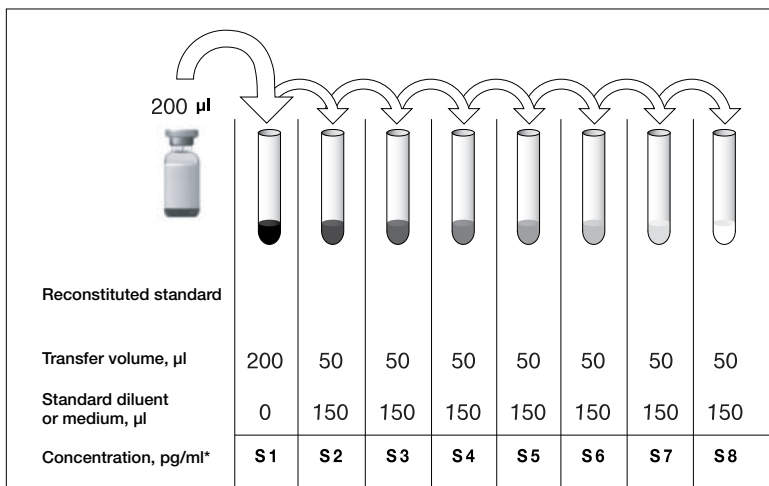
1. Label a set of 1.5 ml Eppendorf tubes as shown in the diagram on the next page.
2. Pipet the appropriate volume of standard diluent into the tubes (see diagram below).

- Transfer the reconstituted standards vial to the first 1.5 ml tube labeled "S1." Vortex gently. This is identified as S1 in the diagram below and in the product insert provided with assay.
- Continue making serial dilutions of the standard as shown. After making each dilution, vortex gently. Change the pipet tip after every transfer.

Note: Running an additional two 0 pg/ml blanks is strongly recommended. Use 50 μ l of the standard diluent as the blank sample. The 0 pg/ml points should be formatted as blanks, not as points in the curve, when using Bio-Plex Manager Software. The blank wells are also useful for troubleshooting and determining the limit of detection (LOD).

- Keep the standards on ice until ready for use. Standards should be used immediately and should not be frozen for future use.

Standard dilution series.



* Refer to the product data sheet for the S1 value for each analyte. Each standard is a fourfold dilution of the preceding one.

Section 7

Control Preparation (Optional)

One tube of lyophilized cytokine control is provided in each Bio-Plex Pro™ Precision Human Cytokine Assay. The preparation of high, medium, and low controls is optional to monitor plate-to-plate variations. This section provides instructions on how to reconstitute the lyophilized control. The product insert provided with the assay lists the concentration of the reconstituted control. The reconstituted control can then be further diluted to prepare any concentration of user-specified quality controls. To ensure optimal assay performance, the cytokine controls should be prepared in a manner consistent with that used to prepare the cytokine standards.

Reconstitute Cytokine Controls

1. Gently tap the glass vial containing the lyophilized cytokine control on a solid surface to ensure the pellet is at the bottom.
2. Reconstitute 1 vial of lyophilized control with 200 µl of dH₂O. Do not use assay buffer to dilute controls.

Sample types and diluents.

Sample	Diluent
Serum	standard diluent
Plasma	standard diluent
Culture supernatant	Same culture medium used to prepare samples

3. Gently vortex for 1–3 sec and incubate on ice for 30 min. Be consistent with the incubation time to ensure optimal assay performance.
4. The reconstituted cytokine control should be further diluted to create the desired QC samples in the same diluents specified in the table above. To obtain the concentration of each reconstituted cytokine control, refer to the product data sheet provided with the assay.

Section 8

Assay Instructions

The following instructions apply to Bio-Plex Pro™ Precision Human Cytokine Assays. All of the necessary components are provided premixed for ease of use.

Plan Experiment

1. Assign which wells of a 96-well plate will be used for each standard, control, and sample (see the example below).
2. Determine the total number of wells that will be used in the assay. Include 25% excess (or add two wells for every eight wells used) to ensure that enough diluted coupled beads, detection antibodies, and streptavidin-PE are prepared.

Example plate.

	1	2	3	4	5	6	7	8	9	10	11	12
A	1	1	B	B	5	5	13	13	21	21	29	29
B	2	2	1	1	6	6	14	14	22	22	30	30
C	3	3	2	2	7	7	15	15	23	23	31	31
D	4	4	3	3	8	8	16	16	24	24	32	32
E	5	5	1	1	9	9	17	17	25	25	33	33
F	6	6	2	2	10	10	18	18	26	26	34	34
G	7	7	3	3	11	11	19	19	27	27	35	35
H	8	8	4	4	12	12	20	20	28	28	36	36

Prepare Coupled Magnetic Beads

Protect the beads from light by covering the tubes with aluminum foil. Keep all tubes on ice until ready to use.

1. Vortex the coupled beads (20x) at medium speed for 15–20 sec.
2. Prepare a sufficient volume of coupled beads (1x) using assay buffer. Each well requires 2 μ l of coupled beads (20x) adjusted to a final volume of 50 μ l with assay buffer (refer to the example below).

Bead calculation examples.

# of Wells	20x Beads, μ l	Assay Buffer, μ l	Total Volume, μ l
96	300	5,700	6,000
48	150	2,850	3,000
32	100	1,900	2,000
24	75	1,425	1,500

Assay Procedure

Bring all buffers to room temperature. Avoid bubbles when pipetting.

Assay Key. The following terms are repeated throughout the assay procedure. Refer to these detailed instructions when wash and incubate are shown in bold.

Term	Detailed Directions
Wash	Add 100 μ l of assay buffer to each well. Wash using Bio-Plex Pro Wash Station automated wash station or by using a handheld magnetic plate washer. When using the handheld magnetic washer, ensure plate is securely attached; invert plate and shake out volume from wells to wash.
Incubate	Gently cover the plate with a new sheet of sealing tape. Place the plate on a microplate shaker and then cover with aluminum foil. Shake the plate at room temperature at 850 ± 50 rpm for the specified incubation time.

1. Equilibrate the diluted standards, samples, and controls at room temperature for 20 min prior to use.
2. Vortex the coupled magnetic beads (1x) for 10–20 sec at medium speed. Add 50 μ l to each well.
3. Wash the plate two times with 100 μ l per well of the 1x assay buffer.
4. Gently vortex the diluted standards, controls, and samples for 1–3 sec. Add 50 μ l of standard, control, sample, or blank to each well, changing the pipet tip after every volume transfer. Duplicate wells are recommended.
5. Cover plate with plate seal and protect from light with aluminum foil. Incubate on shaker at 850 ± 50 rpm for 1 hr at room temperature
6. After incubating the samples, slowly remove and discard the plate seal.
7. Wash 3 times with 100 μ l assay buffer.
8. Vortex the detection antibodies for 10–20 sec, then add 25 μ l to each well. Cover plate with plate seal and protect from light with aluminium foil. Incubate on shaker at 850 ± 50 rpm for 30 min at room temperature.
9. While the detection antibodies are incubating, perform a 30 sec quick-spin centrifugation of the streptavidin-PE (100x) to collect the entire volume at the bottom of the vial prior to pipetting.
10. Prepare a sufficient volume of streptavidin-PE (1x) using assay buffer. Each well requires 0.5 μ l of streptavidin-PE (100x) adjusted to a final volume of 50 μ l with assay buffer (refer to the example on the following page).

Streptavidin-PE calculation examples.

# of Wells	100x Streptavidin-PE, μ l	Assay Buffer, μ l	Total Volume, μ l
96	60	5,940	6,000
48	30	2,970	3,000
32	20	1,980	2,000
24	15	1,425	1,500

11. After the detection antibody incubation, slowly remove and discard the sealing tape.
12. Wash 3 times with 100 μ l of assay buffer.
13. Vortex the streptavidin-PE (1x) vigorously and add 50 μ l to each well. Cover and incubate at 850 ± 50 rpm for 10 min at room temperature.
14. After the streptavidin-PE incubation, slowly remove and discard the sealing tape.
15. Wash 3 times with 100 μ l of assay buffer.
16. Add 100 μ l of assay buffer to each well. Incubate for 30 sec to resuspend the beads. Acquire the data immediately as described in Section 9.

Section 9

Data Acquisition

Bio-Plex Pro™ Precision Cytokine Assays require the use of Bio-Plex Manager™ software version 4.1 or higher. Recommendations for acquiring data using the Bio-Plex® Suspension Array System are listed below. Alternatively, refer to the Bio-Plex Manager™ Software user guide or the instructions provided with the Luminex instrument.

Prepare System

1. Empty the waste bottle and fill the sheath fluid bottle before starting (if HTF not present). This will prevent fluidic system backup and potential data loss.
2. Turn on the reader and microplate platform (and HTF if present). Allow the system to warm up for 30 min.
3. Select **Start up**  and follow the instructions to prepare the reader to acquire data. If the system is idle for 4 hr, the lasers will automatically turn off and a 30 min warm-up period will again be required prior to acquiring data. Select **Warm up**  and wait for the lasers optics to reach operational temperature.

Calibrate System

Calibrate using Bio-Plex Calibration Beads and target values. Daily calibration is recommended before acquiring data.

1. Select **Calibrate**  and confirm that the default values for CAL1 and CAL2 are the same as the values on the Bio-Plex Calibration Bead labels. Use the Bio-Plex Low RP1 target value.
2. Select **OK** and follow the instructions for CAL1 and CAL2 calibration.

Prepare Protocol

1. Open a new protocol by selecting **File > New** from the main menu. Locate the steps at the left of the protocol menu.

Note: To minimize data entry, go to bio-rad.com/web/bio-plex to download preset lot-specific Bio-Plex Pro Precision Human Cytokine Assay protocols.

2. Select step 1 (**Describe Protocol**) and enter information about the assay.
3. Select step 2 (**Select Analytes**) and choose the panel for Cytokines. Choose the target proteins for the assays on the plate. Note that this information will already be entered with the preset downloaded protocol.

Plate formatting example.

The screenshot displays the 'Plate Formatting' tab in the software. The main area shows a 12x8 grid of wells (A1-H12). Wells A1, A2, B1, and B2 are highlighted with diamond shapes, indicating they are selected for standard formatting. The grid contains numerical values for each well, representing the assay results.

	1	2	3	4	5	6	7	8	9	10	11	12
A	1	1	B	B	5	5	6	6	7	7	8	8
B	2	2	1	1	9	9	10	10	11	11	12	12
C	3	3	2	2	13	13	14	14	15	15	16	16
D	4	4	3	3	17	17	18	18	19	19	20	20
E	5	5	1	1	21	21	22	22	23	23	24	24
F	6	6	2	2	25	25	26	26	27	27	28	28
G	7	7	3	3	29	29	30	30	31	31	32	32
H	8	8	4	4	33	33	34	34	35	35	36	36

4. Select step 3 (**Format Plate**) and click the Plate Formatting tab. Click on **S** and drag the cursor over all the wells that contain standards.

Then click on  and drag the cursor over the wells that contain blanks. Repeat with  to identify all the wells that contain controls and  to identify all the wells that contain samples.

Note: If the preset protocol was downloaded, a formatted plate will already be provided. Make any necessary changes to the preset formatted plate so that it matches your plate setup.

5. Select step 4 (**Enter Standards Info**) to enter standards information. Note that this information will already be entered with the preset download protocol.
 - a. Select each analyte individually from the dropdown list.
 - b. Select Enter Automatically and then select the most concentrated value as S1.
 - c. Enter the concentration of S1 from the product insert provided with the assay.
 - d. Enter the dilution factor as 4 and select **Calculate**. The standards information will be populated for the selected analyte.
 - e. Deselect **Apply same concentrations to all analytes**. Repeat steps 5a through 5d for each analyte in the assay.
6. Select Step 5 (**Enter Controls Info**) to enter controls information. This is where the concentration of the user-specified controls is entered into the protocol.
 - a. Select each analyte individually from the dropdown list.
 - b. Enter the description, concentration, and dilution information for each user-specified control.
 - c. Deselect **Apply same concentrations to all analytes**. Repeat steps 6a and 6b for each analyte in the assay.
7. Select step 6 (**Enter Sample Info**) and enter sample information.

Acquire Data

1. Shake the assay plate at 850 ± 50 rpm for 30 sec immediately before acquiring data. Failure to do so will increase data acquisition time due to bead settling.
2. Visually inspect the plate and ensure that the assay wells are filled with buffer prior to placing the plate in the Bio-Plex microplate platform.
3. Slowly remove the sealing tape and any plate cover before placing the plate in the reader.
4. Select step 7 (**Run Protocol**):

- a. Specify data acquisition for 100 beads per region.
- b. In Advanced Settings, set the Bead Map to 25 region.

Note: Bio-Plex Pro Precision Human Cytokine Assays contain magnetic beads and require the use of the 25 region map available in Bio-Plex Manager software version 4.1 or higher.

- c. In Advanced Settings, set the sample size to 50 μ l.
- d. In Advanced Settings, confirm that the default DD gate values are set to 5000 (low) and 32000 (high).

Note: When using a Luminex instrument, set the gates according to the procedure in the Luminex manual.

- e. Select **Start** and save the .rbx file. Then follow the instructions for data acquisition.
5. If acquiring data from more than one plate, empty the waste bottle and refill the sheath bottle after each plate (if HTF not present). Select **Wash Between Plates**  and follow the instructions for fluidics maintenance. Then repeat the **Prepare Protocol** and **Acquire Data** steps.

Note: Use the **Wash Between Plates** command after every plate run to reduce the possibility of the instrument clogging.

6. When data acquisition is complete, select **Shut down**  and follow the instructions.

Reacquire Data

It is possible to acquire data from a well or plate a second time using the Rerun/Recovery mode located below Start in Step 7 (Run Protocol).

1. Check the wells where data will be acquired a second time. Any previous data will be overwritten.
2. Aspirate assay buffer with the wash method of choice, but do not perform the wash step.
3. Add 100 µl of assay buffer to each well. Cover the plate with a new sheet of sealing tape.
4. Repeat **Acquire Data** steps 1–6 to acquire data a second time. The data acquired should be similar to the data acquired initially. However, the data acquisition time will be extended since fewer beads are present in each well.

Section 10

Troubleshooting Guides

This troubleshooting guide addresses problems that may be encountered with Bio-Plex Pro™ Precision Human Cytokine Assays. If you experience any of the problems listed below, review the possible causes and solutions provided. This will assist you in resolving problems directly related to how the assay steps should be performed. Poor assay performance may also be due to the Bio-Plex® Array Reader. To eliminate this possibility, we highly recommend use of the Bio-Plex Validation Kit. This kit will validate all the key functions of the array reader and assist the user in determining whether the array reader is functioning properly.

Possible Causes	Possible Solutions
High Inter-Assay CV	
Standards were not reconstituted consistently	Always be consistent with the incubation time and temperature.
Reconstituted standards and diluted samples were not stored properly	Reconstituted standards and diluted samples should be prepared on ice as instructed. Prior to plating, the reconstituted standards and diluted samples should be equilibrated to room temperature.
Pipetting technique	Pipet carefully and slowly when adding standards, samples, detection antibodies, and streptavidin-PE, especially when using a multichannel pipet. Use a calibrated pipet. Change pipet tip after every volume transfer.

Possible Causes	Possible Solutions
Reagents and assay components were not equilibrated to room temperature prior to plating	All reagents and assay components should be equilibrated to room temperature prior to plating.
Contamination with wash buffer during wash steps	During the wash steps, be careful not to splash wash buffer from one well to another. Be sure that the wells are filtered completely and that no residual volume remains. Also, be sure that the microplate shaker setting is not too high. Reduce the microplate shaker speed to minimize splashing.
Slow pipetting of samples and reagents across the plate	Sample pipetting across the entire plate should take less than 4 min. Reagent pipetting across the entire plate should take less than 1 min.

Low Bead Count

Miscalculated bead dilution	Check your calculations and be careful to add the correct volumes.
Beads clumped in multiplex bead stock tube	Vortex for 15–20 sec at medium speed before aliquoting beads.
Did not shake plate enough before incubation steps and prior to reading	Shake the plate at 850 ± 50 rpm for 30 sec before incubation steps and immediately before reading the plate.
Reader is clogged	Refer to the troubleshooting guide in the Bio-Plex hardware instruction manual.

Possible Causes	Possible Solutions
Low Signal or Poor Sensitivity	
Standards reconstituted incorrectly	Follow the cytokine standard instructions carefully.
Detection antibody or streptavidin-PE diluted incorrectly	Check your calculations and be careful to add the correct volumes.
High Background Signal	
Incorrect buffer was used (for example, assay buffer used to dilute standards)	Use sample matrix or serum standard diluent to dilute cytokine standards.
Spiked "0 pg/ml" wells by mistake	Be careful when spiking standards. Do not add any antigens in the 0 (blank) point.
Streptavidin-PE incubated too long	Follow the procedure incubation time.
Poor Recovery	
Expired Bio-Plex Reagents were used	Check that reagents have not expired. Use new or unexpired components.
Incorrect amounts of components were added	Check your calculations and be careful to add the correct volumes.
Microplate shaker set to an incorrect speed	Check the microplate shaker speed and use the recommended setting. Setting the speed too high may cause splashing and contamination. Use the recommended plate shaker.

Possible Causes	Possible Solutions
Pipetting technique	Pipet carefully and slowly when adding standards, samples, detection antibodies, and streptavidin-PE, especially when using a multichannel pipet. Use a calibrated pipet. Change pipet tip after every volume transfer.

Section 11

Safety Considerations

Eye protection and gloves are recommended while using this product. Consult the SDS for additional information.

Human Source Material — Treat as Potentially Infectious

The Bio-Plex Pro™ Precision Human Cytokine Assays contain components of human origin. This material should be handled as if capable of transmitting infectious agents. Please use universal precautions. The material has been tested by an FDA-approved test and found negative for HBsAg, HIV 1/2 Ab, HIV-1 Ag, and HCV. No test method can provide total assurance that hepatitis B virus, hepatitis C virus, human immunodeficiency virus, or other infectious agents are absent. These components should be handled at Biosafety Level 2 containment [U.S. Government publication: Biosafety in Microbiological and Biomedical Laboratories (CDC, 1999)]. Handle Bio-Plex Pro Precision serum and plasma standard diluents as potentially biohazardous material under at least Biosafety Level 2 containment.

Costar is a trademark of Corning Inc. Eppendorf is a trademark of Eppendorf-Netheler-Hinz GmbH. Luminex and xMAP are trademarks of Luminex Corporation. Vortex-Genie is a trademark of Scientific Industries, Inc.

The Bio-Plex Suspension Array System includes fluorescently labeled microspheres and instrumentation licensed to Bio-Rad Laboratories, Inc. by the Luminex Corporation.

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